United States Patent	12399289
Kind Code	B2
Date of Patent	August 26, 2025
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Radiation detector and nuclear medicine diagnostic apparatus

Abstract

A radiation detector includes a sensor, an A/D converter, and processing circuitry. The sensor outputs a pulse signal based on incidence of radiation. The A/D converter generates digital data by performing A/D conversion on the pulse signal. The processing circuitry acquires temperature information of the A/D converter and outputs digital data that is compensated based on the temperature information.

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Appl. No.: **18/358376**

Filed: **July 25, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240077627 A1	Mar. 07, 2024

Foreign Application Priority Data

JP	2022-118372	Jul. 26, 2022

Publication Classification

Int. Cl.: **G01T1/29** (20060101); **A61B6/03** (20060101); **A61B6/42** (20240101); **G01T1/24** (20060101)

U.S. Cl.:

CPC **G01T1/2985** (20130101); **A61B6/037** (20130101); **A61B6/4241** (20130101); **G01T1/247** (20130101); **G01T1/249** (20130101); G01T1/248 (20130101)

Field of Classification Search

CPC: A61B (6/037); A61B (6/4241); A61B (6/4488); G01T (1/247); G01T (1/248); G01T (1/249); G01T (1/2985); G01T (1/17)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-118372, filed on Jul. 26, 2022, the entire contents of which are incorporated herein by reference.
FIELD
(2) Embodiments described herein relate generally to a radiation detector and a nuclear medicine diagnostic apparatus.
BACKGROUND
(3) Silicon photo multiplier (SiPM) based radiation detectors are known as radiation detectors of nuclear medical diagnosis apparatuses, such as a positron emission computed tomography (PET) apparatus, or radiation diagnostic apparatuses, such as a photon counting computed tomography (PCCT) apparatus.
(4) Analog signal processing in a radiation detector is known as being affected by temperature. For this reason, there has been a problem in that, when a temperature change occurs in the radiation detector due to an effect of unevenness in heat radiation or cooling, a change in surrounding environments, or the like, an output value of energy information or time information changes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating an example of a configuration of a positron emission computed tomography (PET) apparatus according to an embodiment;

(2) FIG. **2** is a diagram illustrating an example of a detector unit according to the embodiment;

(3) FIG. **3** is a diagram for describing temperature compensation according to the embodiment;

(4) FIG. **4** is a diagram for describing the temperature compensation according to the embodiment;

(5) FIG. **5** is a diagram for describing the temperature compensation according to the embodiment; and

(6) FIG. **6** is a flowchart illustrating an example of a flow of an output control process according to the embodiment.

DETAILED DESCRIPTION

(7) A radiation detector according to an embodiment includes a sensor, an A/D converter, and processing circuitry. The sensor outputs a pulse signal based on incidence of radiation. The A/D converter generates digital data by performing A/D conversion on the pulse signal. The processing circuitry acquires temperature information of the A/D converter and outputs digital data that is compensated based on the temperature information.

(8) A radiation detector and a nuclear medicine diagnostic apparatus according to each embodiment will be described with reference to the drawings. In the following description, components having the same or approximately the same functions as those described with respect to the drawings previously presented are denoted with the same reference numerals and are described redundantly only when necessary. Even when the same part is presented, mutual sizes and ratios are sometimes presented differently depending on the drawings. For example, in view of ensuring visibility of the drawings, only main or representative components are sometimes denoted with reference numerals in the description of each drawing and even components having the same or approximately the same functions sometimes are not denoted with reference numerals.

First Embodiment

(9) FIG. **1** is a diagram illustrating an example of a configuration of a positron emission computed tomography (PET) apparatus **100** according to a first embodiment. For example, as illustrated in FIG. **1**, the PET apparatus **100** includes a gantry apparatus **10** and a console apparatus **20**.

(10) The gantry apparatus **10** detects annihilation gamma rays that are emitted because of annihilation of a positron that is emitted from a tracer that is applied to a subject P with an electron and counts the detected annihilation gamma rays, thereby collecting count information. The gantry apparatus **10** has a cylindrical opening that is formed such that the opening penetrates the gantry apparatus **10** in a horizontal direction. The gantry apparatus **10** detects annihilation gamma rays that are emitted from the subject P that is arranged in the opening. Note that, in the following description, the direction along the axis of the cylindrical opening of the gantry apparatus **10** is defined as a Z-axis direction. A horizontal direction orthogonal to the Z-axis direction is defined as an X-axis direction. A vertical direction orthogonal to the Z-axis direction and the X-axis direction is defined as a Y-axis direction.

(11) Specifically, the gantry apparatus includes a tabletop **11**, a table **12**, a table drive mechanism **13**, and a PET detector **14**.

(12) The tabletop **11** is a bed on which the subject is laid. For example, the tabletop **11** is formed into a form of a rectangular flat board and is arranged such that its longitudinal direction is parallel with the Z-axis direction.

(13) The table **12** supports the tabletop **11** such that the tabletop **11** is movable in the X-axis direction, the U-axis direction, and the Y-axis direction.

(14) The table drive mechanism **13** is provided inside or outside the table **12** and causes the tabletop **11** that is supported by the table **12** to move. For example, to capture an image of the subject P, the table drive mechanism **13** causes the tabletop **11** with the subject P being laid thereon

to move into the opening of the gantry apparatus **10**. For example, the table drive mechanism **13** causes the tabletop **11** to move on the table **12** with the position of the table **12** being fixed. Alternatively, for example, the table drive mechanism **13** may include a move base ad cause the tabletop **11** to move together with the table **12** on the move base.
(15) The PET detector **14** detects annihilation gamma rays that are emitted from the subject P. The PET detector **14** generates count information containing detection positions in which the detected annihilation gamma rays are detected, an energy value, and a time of detection and transmits the generated count information to the console apparatus **20**.
(16) Specifically, the PET detector **14** includes a plurality of detector units **14***a* that are arrayed in a form of a ring about the Z-axis such that the detector units **14***a* surround the opening that is formed in the gantry apparatus **10** and each of the detector units **14***a* detects an annihilation gamma ray and generates count information.
(17) The detector unit **14***a* is for example, an anger-type detector employing a photon-counting system and includes a scintillator, a photodetector device and a light guide.
(18) The scintillator converts the annihilation gamma ray that is emitted from the positron in the subject P and that is incident on the scintillator into scintillation light and outputs the scintillation light. For example, the scintillator is formed of scintillator crystals of LaBr3, LYSO, LSO, LGSO, BGO, GAGG, LuAG, or the like, suitable for energy measurement. For example, the scintillator has a two-dimensional array.
(19) The photodetector device detects the scintillation light that is output from the scintillator and converts the scintillation light into an analog signal. The photodetector device consists of, for example, a silicon photomultiplier (SiPM); however, the photodetector device may consist of another photomultiplier tube, such as a photomultiplier (PMT).
(20) The light guide is formed of a plastic material with superior optical transmissivity and transmits the scintillation light that is output from the scintillator to the photodetector device.
(21) The detector unit **14***a* may be a non-anger-type detector without a light guide in which a scintillator and a photodetector device are optically joined in a pair. Alternatively, for example, the detector unit **14***a* may be not an indirect-conversion-type detector using a scintillator but a direct-conversion-type detector using a semiconductor, such as CZT, CdTe, Ge, or Si.
(22) Based on an analog signal that is output from the photodetector device, the detector unit **14***a* generates count information containing a position of detection of an annihilation gamma ray, an energy value, and a time of detection.
(23) For example, the detector unit **14***a* specifies a plurality of photodetector devices that convert scintillation light into analog signals at the same timing. The detector unit **14***a* specifies a position in the scintillator where an annihilation gamma ray is incident as a position of detection of the annihilation gamma ray. For example, by performing center-of-gravity computation based on the position of the photodetector device and the intensity of the analog signal, the detector unit **14***a* specifies the position in the scintillator where the annihilation gamma ray is incident. Alternatively, for example, when the device sizes of the scintillator and the photodetector device correspond to each other, the detector unit **14***a* may specify the position in the scintillator corresponding to the photodetector device from which an output is obtained as a position in the scintillator where the annihilation gamma ray is incident.
(24) For example, the detector unit **14***a* performs integral computation on the intensity of the analog signal that is output from the photodetector device, thereby specifying the energy value of the annihilation gamma ray. Alternatively, for example, the detector unit **14***a* may measure a time (time over threshold (ToT)) in which the intensity of the analog signal that is output from the photodetector device exceeds a threshold that is set in advance and, by performing non-linear correction using the measured time, measure the energy value of the annihilation gamma ray.
(25) For example, the detector unit **14***a* specifies the time in which the scintillation light is detected by the photodetector device as a time of detection of the annihilation gamma ray. The time of

detection may be an absolute time or an elapsed time from the time of start of imaging.
(26) The console apparatus **20** receives various types of operations on the PET apparatus **100** from an operator and controls operations of the PET apparatus **100** according to the received operations. Specifically, the console apparatus **20** includes an input interface **21**, a display **22**, a memory **23**, and a processing circuitry **24**. Each unit that the console apparatus **20** includes is connected via a bus. A case where the gantry apparatus **10** and the console apparatus **20** are independent from each other is exemplified here; however, the gantry apparatus **10** may include the console apparatus **20** or part of the components of the console apparatus **20**.
(27) The input interface **21** receives various input operations from the operator, converts the received input operations into electric signals, and outputs the electric signals to the processing circuitry **24**. For example, the input interface **21** receives operational inputs for setting imaging conditions and a region of interest (ROI) from the operator.
(28) For example, a mouse, a keyboard, a trackball, a switch, a button, a joystick, non-contact input circuitry using an optical sensor, audio input circuitry, a touch pad, a touch panel display, and the like, are usable as appropriate as the input interface **21**.
(29) In the first embodiment, the input interface **21** is not limited to one including these physical operational parts. For example, examples of the input interface **21** include an electric signal processing circuitry that receives an electric signal corresponding to an input operation from an external input device that is provided independently of the apparatus and that outputs the electric signal to the processing circuitry **24**. The input interface **21** may be provided in the gantry apparatus **10**. The input interface **21** may consist of a tablet terminal device capable of radio communication with the console apparatus **20**. The input interface **21** is an example of the input unit.
(30) The display **22** displays various types of information. For example, the display **22** outputs a medical image (a PET image) that is generated by the processing circuitry **24** and a graphical user interface (GUI) for receiving various types of operations from the operator. Various freely-selected displays are usable as appropriate as the display **22**. For example, a liquid crystal display (LCD), a cathode ray tube (CRT) display, an organic electro luminescence display (OELD), or a plasma display is usable as the display **22**.
(31) Note that the display **22** may be provided anywhere in a control room. The display **22** may be provided on the gantry apparatus **10**. The display **22** may be a desktop-type one or may consist of a tablet terminal device capable of radio communication with the main unit of the console apparatus **20**. At least one projector may be used as the display **22**. The display **22** is an example of a display unit.
(32) The memory **23** stores various types of data and programs that are used in the PET apparatus **100**. The memory **23** is realized by, for example, a semiconductor memory device, such as a read only memory (ROM), a random access memory (RAM), or a flash memory, a hard disk, or an optical disk. The storage area of the memory **23** may be in the PET apparatus **100** or may be in an external storage device that is connected via a network. The memory **23** is an example of a storage unit.
(33) The processing circuitry **24** controls entire operations of the PET apparatus **100**. The processing circuitry **24** includes a processor and a memory, such as a ROM or a RAM, as hardware resources. The processing circuitry **24** executes various types of functions, such as a controlling function **24***a*, a coincidence count information generating function **24***b*, and an image reconstructing function **24***c*, using a processor that executes a program that is loaded in the memory. The processing circuitry **24** is an example of a processor.
(34) In the controlling function **24***a*, the processing circuitry **24** controls each unit of the gantry apparatus **10** and the console apparatus **20**, thereby controlling the entire PET apparatus **100**. For example, the processing circuitry **24** controls the table drive mechanism **13** to move the tabletop **11**. For example, the processing circuitry **24** controls the PET detector **14** to collect count information

and stores the collected count information in the memory **23**.

(35) In the coincidence count information generating function **24***b*, the processing circuitry **24** generates coincidence count information using the count information that is collected. Specifically, the processing circuitry **24** refers to the count information that is stored in the memory **23** and, based on the times of detection of the respective sets of count information, specifies a pair of sets of count information obtained by counting annihilation gamma rays approximately concurrently. The processing circuitry **24** generates coincidence count information in which the specified sets of count information in a pair are associated with each other and stores the generated coincidence count information in the memory **23**. Based on a plurality of sets of time information that are acquired by the detector units **14***a*, the processing circuitry **24** corrects a difference in time information between the detector unit **14***a* serving as a reference and the detector unit **14***a* that is another detector unit.

(36) In the image reconstructing function **24***c*, the processing circuitry **24** reconstructs a PET image based on the coincidence count information that is generated by the coincidence count information generating function **24***b*. Specifically, the processing circuitry **24** reads the coincidence count information that is stored in the memory **23** and performs back projection processing using the read coincidence count information as projection data, thereby reconstructing a PET image. The processing circuitry **24** stores the reconstructed PET image in the memory **23**.

(37) Each of the functions **24***a* to **24***c* is not limited to the case where the functions are realized by a single set of processing circuitry. A plurality of independent processors may be combined to configure the processing circuitry **24** and the processors may execute programs, respectively, thereby implementing the respective functions **24***a* to **24***c*. The functions **24***a* to **24***c* may be distributed or integrated as appropriate to a plurality of sets of processing circuitry or into a single processing circuitry.

(38) Note that the console apparatus **20** is described as one that executes the functions using a single console; however, different consoles may execute the functions. For example, the console apparatus **20** may have the functions of the processing circuitry **24**, such as the coincidence count information generating function **24***b* and the image reconstructing function **24***c*, in a distributed manner.

(39) Part of or all the processing circuitry **24** is not limited to the case where it is included in the console apparatus **20** and part of or all the processing circuitry **24** may be included in an integration server that collectively performs processing on data that is acquired in a plurality of medical image diagnostic apparatuses.

(40) At least one of a process of generating coincidence count information and a process of reconstructing a PET image may be performed by any one of the console apparatus **20** and an external work station. Both the console apparatus **20** and the work station may perform the processes concurrently. For example, a computer including a processor that implements functions corresponding to the respective processes and a memory, such as a ROM or a RAM, as hardware resources is usable as appropriate as the work station.

(41) A configuration example of the PET apparatus **100** according to the first embodiment has been described. Under such a configuration, as described above, the PET apparatus **100** specifies a pair of sets of count information obtained by counting annihilation gamma rays approximately concurrently from the count information on annihilation gamma rays that are detected by the detector units **14***a* and, based on the coincidence count information in which the specified sets of count information in a pair are associated with each other, generates a PET image.

(42) In the PET apparatus **100** described above, analog signal processing in the PET detector **14** is affected by temperature. There is a possibility that a temperature change in the PET detector **14** lowers the timing of detection and an energy resolution. For this reason, it is important to put the PET detector **14** in a stable temperature environment.

(43) Photodetector devices (photoelectric conversion devices), such as a SiPM, and A/D conversion

devices are known as particularly tending to be affected by temperature. More specifically, as for the effect of temperature on the PET detector **14**, the effect on the photoelectric conversion device is dominant. For this reason, in general, the voltage to be applied is varied according to a change in the temperature of the photoelectric conversion device and a compensation is made in order to obtain stable energy information. Specifically, a parameter is adjusted in order to perform appropriate voltage control based on the result of actual measurement of the temperature of the photoelectric conversion device.

(44) There is however a risk that only the temperature of the A/D conversion device changes in the case where an abrupt temperature change occurs in the PET detector **14**, or the like. In this case, there is a problem in that, when the temperature of the A/D conversion device changes, output values of the energy information and the time information change abruptly according to a resolution of the A/D conversion device.

(45) Thus, the PET detector **14** according to the first embodiment is configured to output digital data on which a temperature compensation based on the temperature of an A/D converter **145** is made. The PET detector **14** is an example of a radiation detector.

(46) A temperature compensation further based on the temperature of the photoelectric conversion device in addition to the temperature of the A/D converter **145** can be made.

(47) The configuration example of the PET detector **14** according to the first embodiment will be described more in detail. FIG. **2** is a diagram illustrating a configuration example of the detector unit **14***a* according to the first embodiment.

(48) In the PET detector **14**, each of the detector units **14***a* includes an analog unit **141**, a data acquisition system (DAS) and a temperature sensor **147**. For example, the DAS includes an oscillator **142**, a phase locked loop (PLL) **143**, a divider **144**, the A/D converter **145**, and application specific integrated circuit (ASIC) **146**.

(49) In the PET detector **14**, event data is collected in each block containing at least one single detector unit **14***a*. In the first embodiment, the case where event data is collected in each detector unit **14***a* will be exemplified.

(50) The analog unit **141** outputs a pulse signal based on incidence of radiation. Specifically, the analog unit **141** detects an analog signal based on the result of detection of a gamma ray. Specifically, the analog unit **141** includes the scintillator and the photodetector device (photoelectric conversion device) described above, converts an annihilation gamma ray that is emitted from a positron in the subject P and is incident into scintillation light, converts the scintillation light into an analog pulse signal, and outputs the analog pulse signal. The pulse signal is referred to as a detection signal below. The analog unit **141** is an example of a sensor unit (sensor).

(51) The oscillator **142** generates a clock signal. For example, the oscillator **142** is realized using circuitry using a built-in oscillator, such as a crystal oscillator.

(52) The phase locked loop (PLL) **143** converts a clock signal that is generated by the oscillator **142** into a clock signal of a pre-determined frequency and outputs the clock signal.

(53) The divider **144** divides the clock signal that is output from the PLL **143** to the A/D converter **145** and the ASIC **146**.

(54) The A/D converter **145** performs A/D conversion on the detection signal from the analog unit **141**, thereby generating digital data. The A/D converter **145** generates digital data (event data) per event. Specifically, on receiving an input of the detection signal that is analog data, the A/D converter **145** converts the detection signal into digital data. The digital data contains the position of detection of annihilation radiation (for example, identification information of the detector unit **14***a*, the photoelectric conversion device or the scintillator), the energy value (for example, the intensity of the detection signal), and the time of detection (for example, the absolute time or the elapsed time from the start of imaging). Based on the clock signal that is generated by the oscillator **142**, the A/D converter **145** measures a time when the intensity of the analog detection signal that is

detected by the analog unit **141** exceeds the given threshold and converts the time into a digital signal, thereby generating the time information. The A/D converter **145** is an example of an A/D converter.

(55) The ASIC **146** is configured to be able to execute various types of functions, such as a temperature acquiring function **146***a*, a time information generating function **146***b*, a count information generating function **146***c*, and an output controlling function **146***d*.

(56) In the temperature acquiring function **146***a*, the ASIC **146** acquires temperature information from the temperature sensor **147**. The ASIC **146** that realizes the temperature acquiring function **146***a* is an example of an acquisition unit.

(57) For example, the temperature information is information presenting the value of the temperature of the A/D converter **145** or the vicinity of the A/D converter **145** that is measured by the temperature sensor **147**. Alternatively, for example, the temperature information may be information presenting a section of the temperature of the A/D converter **145** or the vicinity of the A/D converter **145**. The section of temperature can be set, for example, according to a threshold described below.

(58) In the time information generating function **146***b*, the ASIC **146** acquires the time information (digital data) that is generated by the A/D converter **145** according to a synchronization signal that is generated by a synchronization signal generator. The ASIC **146** transmits the acquired time information to the console apparatus **20**.

(59) For example, the synchronization signal generator may be the scintillator included in the analog unit **141**. The synchronization signal is a gamma ray that is generated by self-decay of the scintillator. For example, the ASIC **146** acquires the time information in response to occurrence of a gamma ray because of self-decay of the scintillator. For example, the synchronization signal generator may be a gamma ray source that is arranged in the opening of the gantry apparatus **10**. The synchronization signal may be a gamma ray that is emitted from the gamma ray. For example, the synchronization signal generator may be the controlling function **24***a* of the console apparatus **20**. The synchronization signal may be a command that is transmitted from the controlling function **24***a*.

(60) In the count information generating function **146***c*, based on the analog detection signal that is output from the analog unit **141** and the time information that is generated by the A/D converter **145**, the ASIC **146** generates the count information (digital data) containing the position of detection of the annihilation gamma ray, the energy value, and the time of detection. The ASIC **146** transmits the generated count information to the console apparatus **20**.

(61) In the output controlling function **146***d*, the ASIC **146** performs output control of outputting the digital data that is compensated based on the temperature information that is acquired from the temperature sensor **147**. Details of the output control based on the temperature information will be described below. The ASIC **146** that realizes the output controlling function **146***d* is an example of an output controller.

(62) The example using the ASIC **146** has been described; however, the processing circuitry is not limited thereto. For example, another set of processing circuitry that is realized by, instead of the ASIC **146**, a processor, such as a central processing unit (CPU), a graphics processing unit (GPU), a field programmable gate array (FPGA), a simple programmable logic device (SPLD), or a complex programmable logic device (CPLD), may be used.

(63) The temperature sensor **147** includes at least a sensor capable of measuring the temperature of the A/D converter **145**. The temperature sensor **147** may be a sensor capable of measuring a temperature of a board on which the A/D converter **145** is mounted as the temperature of the A/D converter **145**. The temperature sensor **147** may further include a sensor capable of measuring a temperature of a SiPM (photoelectric conversion device). For example, a thermistor is used as the temperature sensor **147**, and another sensor capable of measuring a temperature, such as a thermocouple or a radiation thermometer, may be used.

(64) The temperature compensation will be described here more in details with reference to the accompanying drawings. Each of FIGS. **3** to **5** is a diagram for describing temperature compensation according to the first embodiment.
(65) In the output controlling function **146***d*, the ASIC **146** performs output control of setting a threshold for the pulse signal from the analog unit **141** based on the temperature information. In the first embodiment, the threshold for the pulse signal from the analog unit **141** is a threshold for energy measurement on the pulse signal.
(66) FIG. **3** exemplifies an analog waveform W of the detection signal from the A/D converter **145** in a standard temperature environment. When the temperature information is information presenting the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** performs output control of setting a threshold Th**0** for the detection signal from the analog unit **141**. The threshold Th**0** is determined in advance according to a base voltage VB**0** of the A/D converter **145**. The description will be continued, assuming that energy information E**0** is obtained as a result of A/D conversion on the detection signal of the analog waveform W using the threshold Th**0**.
(67) FIG. **4** exemplifies an analog waveform W of the detection signal from the A/D converter **145** at the time when only the temperature of the A/D converter **145** increases. The base voltage of the A/D converter **145** at that time is a voltage VB**1** that is higher than the voltage VB**0** in association with the increase in temperature. For this reason, as illustrated in FIG. **4**, with the increase of the base voltage of the A/D converter **145**, the analog waveform W of the detection signal from the A/D converter **145** transitions to a higher value (upward in FIG. **3** and FIG. **4**). Accordingly, the A/D conversion using the threshold Th**0** corresponding to the standard temperature environment generates energy information E**11** higher than the energy information E**0**.
(68) As described above, because the temperature change occurring in the A/D converter **145** causes a change in the output value of the energy information, the output control based on the temperature information according to the first embodiment changes the threshold that is set for the analog detection signal for performing A/D conversion according to the temperature information.
(69) Specifically, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** performs output control of setting a threshold Th**1** for the detection signal from the analog unit **141**. In other words, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, the ASIC **146** changes the threshold that is set for the analog detection signal for performing A/D conversion from the threshold Th**0** corresponding to the standard temperature environment to the threshold Th**1** corresponding to a high-temperature environment. The threshold Th**1** is a value that is determined in advance according to the base voltage VB**1** of the A/D converter **145** or a range thereof and is higher than the threshold Th**0**.
(70) According to the A/D conversion using the threshold Th**1** corresponding to the high-temperature environment, energy information E**12** lower than the energy information E**11** is generated. It is thus possible to inhibit an increase in the output value of the energy information resulting from the temperature increase occurring in the A/D converter **145**.
(71) FIG. **5** exemplifies an analog waveform W at the time when only the temperature of the A/D converter **145** lowers. The base voltage of the A/D converter **145** at that time is a voltage VB**2** that is lower than the voltage VB**0** in association with the decrease in temperature. For this reason, as illustrated in FIG. **5**, in association with the decrease of the base voltage of the A/D converter **145**, the analog waveform W of the detection signal from the A/D converter **145** transitions to a lower value (downward in FIG. **3** and FIG. **5**). Accordingly, the A/D conversion using the threshold Th**0** corresponding to the standard temperature environment generates energy information E**21** lower than the energy information E**0**.
(72) Thus, in the output control based on the temperature information according to the first

embodiment, when the temperature information is information presenting an environment at a temperature lower than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** performs output control of setting a threshold Th**2** for the detection signal from the analog unit **141**. In other words, when the temperature information is information presenting an environment at a temperature lower than that of the standard temperature environment, the ASIC **146** changes the threshold that is set for the analog detection signal for performing A/D conversion from the threshold Th**0** corresponding to the standard temperature environment to the threshold Th**2** corresponding to a low-temperature environment. The threshold Th**2** is a value that is determined in advance according to the base voltage VB**2** of the A/D converter **145** or a range thereof and is lower than the threshold Th**0**.
(73) According to the A/D conversion using the threshold Th**2** corresponding to the low-temperature environment, energy information E**22** higher than the energy information E**21** is generated. It is thus possible to inhibit a decrease in the output value of the energy information resulting from the temperature decrease occurring in the A/D converter **145**.
(74) A flow of the output control based on the temperature information according to the first embodiment will be described more in detail with reference to the accompanying drawings.
(75) FIG. **6** is a flowchart illustrating an example of a flow of the output control process according to the first embodiment.
(76) First of all, in the temperature acquiring function **146***a*, the ASIC **146** acquires the temperature information of the A/D converter **145** from the temperature sensor **147** (S**101**).
(77) In the output controlling function **146***d*, the ASIC **146** sets a threshold for the pulse signal from the analog unit **141** based on the temperature information as described above with reference to FIGS. **3** to **5** (S**102**).
(78) Note that the process from S**101** to S**102** need not necessarily be performed every time. When there is no change in the temperature information, the same threshold is set and therefore setting a threshold may be omitted.
(79) In the time information generating function **146***b*, the ASIC **146** acquires time information that is generated by the A/D converter **145** in a state where a threshold is set based on the temperature information. In the count information generating function **146***c*, the ASIC **146** generates count information based on the time information (S**103**).
(80) Thereafter, in the output controlling function **146***d*, the ASIC **146** outputs digital data on which a temperature compensation is made based on the time information and the count information that are generated, that is, the temperature information of the A/D converter **145** to the console apparatus **20** (S**104**).
(81) As described above, the temperature compensation according to the first embodiment is realized by changing the threshold, which is set for the analog detection signal for performing A/D conversion, according to the temperature. Specifically, in the output controlling function **146***d*, the ASIC **146** sets a threshold for the pulse signal from the analog unit **141** based on the temperature information. In other words, the ASIC **146** performs the output control such that a change in the output value from the A/D converter **145** that is associated with a temperature change is inhibited and digital data that is compensated based on the temperature information is output from the PET detector **14**.
(82) Accordingly, even when there is a significant temperature increase or a temperature decrease in the A/D converter **145** in association with an abrupt change in temperature because of the outside air or high count rate processing, it is possible to minimize fluctuations in energy information.
Second Embodiment
(83) In the first embodiment, the example in which a threshold for energy measurement on the pulse signal from the analog unit **141** is set based on the temperature information is described as an example; however, the threshold is not limited thereto. In the output controlling function **146***d*, the ASIC **146** may set a threshold for the timing of the pulse signal from the analog unit **141** based on

the temperature information.
(84) The threshold for the timing of the pulse signal is a threshold for the intensity of the detection signal from the analog unit **141** that is used by the A/D converter **145** to generate time information.
(85) Even with this configuration, the ASIC **146** is able to inhibit a change in the output value from the A/D converter **145** that is associated with a temperature change.
(86) Note that it is possible to combine the technique according to a second embodiment with the technique according to the first embodiment as appropriate. Specifically, the ASIC **146** may be configured to set each of a threshold for energy measurement on the pulse signal from the analog unit **141** and a threshold for generation of time information from the pulse signal based on the temperature information.
Third Embodiment
(87) In each of the embodiments described above, output control of outputting digital data on which a temperature compensation is made by setting a threshold for the pulse signal from the analog unit **141** is exemplified; however, the output control is not limited thereto. In the output controlling function **146***d*, the ASIC **146** may output digital data on which a temperature compensation is made by controlling operations of the analog unit **141** based on the temperature information.
(88) In the output controlling function **146***d*, the ASIC **146** determines a voltage to be applied to the SiPM (photoelectric conversion device) of the analog unit **141** based on the temperature information. For example, assume that a table or a relation representing the relationship between the temperature information and the voltage to be applied is determined in advance.
(89) Note that the table or the relation may be one presenting correspondence between temperature information of the SiPM and the voltage to be applied. In this case, the temperature sensor **147** further includes a sensor capable of measuring the temperature of the SiPM.
(90) In an example, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** performs output control of applying a voltage lower than that in the case where the temperature information is information presenting the standard temperature environment to the SiPM. In other words, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, the ASIC **146** lowers the voltage to be applied to the SiPM, thereby transitioning the analog waveform W of the detection signal from the analog unit **141** (for example, downward in FIG. **3** and FIG. **4**).
(91) In an example, when the temperature information is information presenting an environment at a temperature lower than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** performs output control of applying a voltage higher than that in the case where the temperature information is information presenting the standard temperature environment to the SiPM. In other words, when the temperature information is information presenting an environment at a temperature lower than that of the standard temperature environment, the ASIC **146** increases the voltage to be applied to the SiPM, thereby transitioning the analog waveform W of the detection signal from the analog unit **141** (for example, upward in FIG. **3** and FIG. **5**).
(92) Also with this configuration, the ASIC **146** is able to inhibit a change in the output value from the PET detector **14** that is associated with a temperature change of the A/D converter **145**.
(93) Note that it is possible to combine the technique according to a third embodiment with the technique according to each of the above-described embodiments as appropriate. For example, the ASIC **146** may be configured to set a threshold for the pulse signal from the analog unit **141** and determine a voltage to be applied to the SiPM based on the temperature information.
Fourth Embodiment
(94) In the output controlling function **146***d* according to a fourth embodiment, the ASIC **146**

corrects time information (digital data) that is generated by the time information generating function **146***b* based on the temperature information. The ASIC **146** the outputs the time information on which the temperature correction is made to the console apparatus **20**.

(95) Also with this configuration, the ASIC **146** is able to inhibit a change in the digital data from the PET detector **14** that is associated with a temperature change of the A/D converter **145**. Note that it is possible to combine the technique according to the fourth embodiment with the technique according to each of the above-described embodiments.

Fifth Embodiment

(96) In the output controlling function **146***d* according to a fifth embodiment, the ASIC **146** corrects the count information (digital data) that is generated by the count information generating function **146***c* based on the temperature information. The ASIC **146** outputs the time information on which the temperature compensation is made to the console apparatus **20**.

(97) Also with this configuration, the ASIC **146** is able to inhibit a change in the digital data from the PET detector **14** that is associated with a temperature change of the A/D converter **145**. It is possible to combine the technique according to the firth embodiment with the technique according to each of the above-described embodiments.

Sixth Embodiment

(98) In the output controlling function **146***d* according to a sixth embodiment, the ASIC **146** outputs a command making an instruction for temperature information or temperature compensation to the console apparatus **20** together with the generated digital data. The command making an instruction for temperature compensation is generated based on the temperature information.

(99) The console apparatus **20** according to the sixth embodiment corrects the digital data from the PET detector **14** based on the command making an instruction for temperature information or temperature compensation from the PET detector **14**. Specifically, in the controlling function **24***a*, the processing circuitry **24** controls the PET detector **14** to collect time information and coefficient information (digital data), makes a temperature compensation on the collected digital data, and stores the digital data after the temperature compensation in the memory **23**. The processing circuitry **24** that realizes the controlling function **24***a* is an example of a controller.

(100) Also with this configuration, in the PET apparatus **100**, it is possible to inhibit the effect of a change of the digital data from the PET detector **14** that is associated with the temperature change of the A/D converter **145**. Note that it is possible to appropriately combine the technique according to the sixth embodiment with the technique according to each of the above-described embodiments.

Seventh Embodiment

(101) In the output controlling function **146***d* according to a seventh embodiment, the ASIC **146** controls a temperature controller that is mounted on the A/D converter **145** based on the temperature information.

(102) The temperature controller is a heater or a cooler that is configured to be able to adjust the temperature of the A/D converter **145** or a combination of the heater and the cooler. For example, a fan is usable as the cooler.

(103) In an example, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** turns the cooler on.

(104) In an example, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** causes the fan to turn at a rotation speed higher than that in the case where the temperature information is information presenting the standard temperature environment.

(105) In an example, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** turns the heater off.

(106) In an example, when the temperature information is information presenting an environment at a temperature lower than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** turns the cooler off.
(107) In an example, when the temperature information is information presenting an environment at a temperature higher than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** causes the fan to turn at a rotation speed lower than that in the case where the temperature information is information presenting the standard temperature environment.
(108) In an example, when the temperature information is information presenting an environment at a temperature lower than that of the standard temperature environment, in the output controlling function **146***d*, the ASIC **146** turns the heater on.
(109) Also with these configurations, the ASIC **146** is able to inhibit a change in the output value from the PET detector **14** that is associated with a temperature change of the A/D converter **145**. It is possible to combine the technique according to the seventh embodiment with the technique of each of the above-described embodiments.
(110) Note that the temperature compensation method according to the embodiment may be realized by an apparatus independent of the PET apparatus **100**, such as a computer including a processor and a memory as hardware resources. In this case, the processor that is mounted on the computer executes a program that is read from the ROM, or the like, and is loaded into the RAM, thereby realizing the temperature compensation method according to the embodiment.
(111) The temperature compensation method according to each of the embodiments may be realized in a photon counting computed tomography (PCCT) apparatus. In this case, a processor that is mounted on the PCCT apparatus executes a program that is read from a ROM, or the like, and is loaded into a RAM, thereby realizing the temperature compensation method according to each of the embodiments.
(112) For example, there are various types of X-ray CT apparatuses including a third-generation CT and a fourth-generation CT and any type is applicable to each of the embodiments. The third-generation CT is a rotate/rotate-type in which an X-ray-tube and a detector rotate as a unit around a subject. In the fourth-generation CT, the fourth-generation CT is a stationary/rotate-type in which a large number of X-ray detection devices that are arrayed in a ring-like form are fixed and only an X-ray tube rotates around a subject.
(113) The word “processor” used in the description given above refers to, for example, circuitry, such as a CPU, a GPU, an ASIC, or a programmable logic device (PLD). The PLD covers a SPLD, a CPLD, and a FPGA. The processor reads programs that are saved in storage circuitry and executes the programs, thereby implementing the functions. The storage circuitry in which the programs are saved is a computer-readable non-temporary recording medium. Instead of saving the programs in the storage circuitry, the programs may be directly installed in the circuitry of the processor. In this case, the processor reads and executes the programs that are installed in the circuitry, thereby implementing the functions. Without executing the programs, the functions corresponding to the programs may be implemented by a combination of sets of logic circuitry. Note that each processor of the embodiments is not limited to the case where each processor is configured as a single set of circuitry, and a plurality of independent sets of circuitry may be combined to configure a single processor to implement the functions. Furthermore, the components in FIG. **1** or FIG. **2** may be integrated into one processor to implement functions thereof.
(114) According to at least one of the embodiments described above, tie is possible to appropriately compensate an effect of a temperature change in a radiation detector.
(115) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their

equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.
(116) With respect to the above-described embodiments, the following appendants are disclosed as one aspect and selective characteristics of the invention.
(117) (Appendant 1)
(118) A radiation detector including: a sensor unit that outputs a pulse signal based on incidence of radiation; an A/D converter that generates digital data by performing A/D conversion on the pulse signal; an acquisition unit that acquires temperature information of the A/D converter; and an output controller that outputs digital data that is compensated based on the temperature information.
(Appendant 2)
(119) A nuclear medicine diagnostic apparatus including: a radiation detector including a sensor unit that outputs a pulse signal based on incidence of radiation, an A/D converter that generates digital data by performing A/D conversion on the pulse signal, an acquisition unit that acquires temperature information of the A/D converter, and an output controller that outputs digital data; and a controller that makes a temperature compensation on the digital data from the radiation detector based on the temperature information.
(Appendant 3)
(120) The output controller may set a threshold for the pulse signal based on the temperature information.
(121) (Appendant 4)
(122) The threshold may be a threshold for energy measurement on the pulse signal.
(123) (Appendant 5)
(124) The threshold may be a threshold for timing of the pulse signal.
(125) (Appendant 6)
(126) The output controller may correct the digital data based on the temperature information.
(127) (Appendant 7)
(128) The digital data may be at least any one of time information and count information.
(129) (Appendant 8)
(130) The output controller may determine a voltage to be applied to the sensor unit based on the temperature information.
(131) (Appendant 9)
(132) The acquisition unit may further acquire temperature information of the sensor unit.
(133) The output controller may determine the voltage to be applied to the sensor unit further based on the temperature information of the sensor unit.
(134) (Appendant 10)
(135) The radiation detector may further include a temperature controller that controls a temperature of the A/D converter.
(136) The output controller may control an operation of the temperature controller based on the temperature information.

Claims

1. A radiation detector comprising: a sensor that outputs a pulse signal based on incidence of radiation; an A/D converter that generates digital data by performing A/D conversion on the pulse signal; and processing circuitry that acquires temperature information of the A/D converter and that outputs digital data that is compensated based on the temperature information.
2. The radiation detector according to claim 1, wherein the processing circuitry sets a threshold for the pulse signal based on the temperature information.
3. The radiation detector according to claim 2, wherein the threshold is a threshold for energy

measurement on the pulse signal.
4. The radiation detector according to claim 2, wherein the threshold is a threshold for timing of the pulse signal.
5. The radiation detector according to claim 1, wherein the processing circuitry corrects the digital data based on the temperature information.
6. The radiation detector according to claim 5, wherein the digital data is at least one of time information and count information.
7. The radiation detector according to claim 1, wherein the processing circuitry determines a voltage to be applied to the sensor unit based on the temperature information.
8. The radiation detector according to claim 7, wherein the processing circuitry further acquires temperature information of the sensor unit, and determines the voltage to be applied to the sensor unit based on the temperature information of the sensor unit.
9. The radiation detector according to claim 1, further comprising a temperature controller that controls a temperature of the A/D converter, and the processing circuitry controls an operation of the temperature controller based on the temperature information.
10. A nuclear medicine diagnostic apparatus comprising: a radiation detector including a sensor that outputs a pulse signal based on incidence of radiation, an A/D converter that generates digital data by performing A/D conversion on the pulse signal, and processing circuitry that acquires temperature information of the A/D converter and that outputs the digital data; and a processor that makes a temperature compensation on the digital data from the radiation detector based on the temperature information.